

Lecture Notes: World Models and the Future of AI

Header Information

Field	Details
Title	World Models and the Future of AI
Instructor	Professor Yann LeCun
Institution	NYU Courant Institute
Event	Applied Mathematics Webinar
Focus	Hierarchical JEPA, Energy-Based Models, and AI Research

Overview

This lecture by Professor Yann LeCun discusses the future of artificial intelligence, with particular emphasis on hierarchical Joint Embedding Predictive Architecture (JEPA), energy-based models, and world models. Professor LeCun explores the mathematical challenges in deep learning, the phenomenon of double descent, and the practical applications of AI in safety-critical industries such as nuclear engineering. The Q&A session addresses the intersection of mathematics and AI research, as well as the deployment of AI systems in complex industrial environments.

Key Points

1. NYU Courant Institute and Applied Mathematics

- Courant Institute is now a standalone school within NYU, separate from law, business, arts, and engineering schools
- The separation reflects the importance of applied mathematics, computer science, and AI/data science
- Professor LeCun maintains connections with the applied mathematics department despite being in computer science
- He served on the science advisory board of the Institute for Pure and Applied Mathematics at UCLA

2. Mathematical Challenges in Deep Learning

The Complexity of Neural Networks

- Neural networks are **non-convex** systems, making traditional mathematical proofs difficult
- Standard optimization theory and criteria do not apply well to neural network training
- **Stochastic gradient descent** behavior is particularly hard to model mathematically

- Traditional optimization tools become ineffective because precise optimization in machine learning leads to overfitting

The Double Descent Phenomenon

- **Traditional statistical theory** states: too many parameters cause overfitting
- Example: Using an 11-degree polynomial to fit 10 data points causes wild behavior between points
- **Empirical reality:** This conventional wisdom is **completely false**
- Neural networks are massively over-parameterized compared to training samples, yet they:
 - Generalize well
 - Interpolate effectively
 - Even extrapolate successfully

Double Descent Definition: As model complexity increases:

1. Error decreases initially (better fit with more capacity)
2. Error increases (traditional overfitting region)
3. Error decreases again at very high complexity (modern neural network behavior)

This phenomenon remains not fully understood theoretically and represents a significant mathematical research opportunity.

3. Energy-Based Models (EBMs)

Why Energy-Based Models?

- Traditional probabilistic modeling requires estimating **joint distributions** between variables
- In high dimensions, distribution estimation is **intractable** (partition function problem)
- The normalization constant in $\exp(-\text{energy})$ is computationally infeasible

The Energy-Based Approach

- Learn a **contrast function** (scalar output) that:
 - Returns **low value (zero)** for data pairs (X, Y) in high-density regions
 - Returns **high value** for pairs away from the data manifold
- Captures dependencies between variables without explicit probability distribution
- Represents a simpler and more practical way to model variable relationships

Key Insight: Energy-based models are a "contrast function" that distinguishes on-manifold data from off-manifold data, avoiding the need for explicit probability estimation.

4. World Models and Hierarchical Planning

World Models vs. Digital Twins

- World models should **not** be confused with digital twins
- World models are phenomenological models trained on data
- They enable planning and prediction for complex systems

JEPA (Joint Embedding Predictive Architecture)

- Enables **hierarchical planning** for intelligent systems
- Goal: Train universal causal models of complex phenomena
- Potential as the foundation for any future intelligent system
- Can process high-dimensional, noisy data (not just tokens like text)

Applications in Safety-Critical Industries

- Control of complex systems (e.g., power plants, nuclear facilities)
- Training approaches:
 1. Real data collection
 2. Synthetic data from digital twin simulations (pre-training)
 3. Completion with real data for realism
 4. Continuous fine-tuning through prediction error correction

5. AI Deployment in Nuclear Engineering

Current Challenges

- Sensory streams and inspection images require robust AI handling
- Maintenance data integration
- Physics-based models and explicit simulators remain important
- **Uncertainty quantification** is mandatory
- Validation and auditability are essential requirements

JEPA-Style Model Integration

- World models as **higher abstraction layers** on top of engineering models
- Planning for anomaly and accident anticipation
- Complementary to (not replacement of) physics-based models
- Connection to Model Predictive Control (MPC) procedures like IDICOM

Small Modular Reactors (SMRs)

- Potential collaboration opportunity with nuclear engineering
- SMRs simplify nuclear power plant construction
- JEPA-style models could provide useful capabilities for SMR monitoring and control

Discussion Points from Q&A

Q1: Mathematics and AI Research

Core Question: Are enough mathematicians dedicating research to AI?

Key Points from Professor LeCun's Response:

- Mathematical problems in AI are "very interesting but also very complicated"
- Traditional mathematicians preferred simpler paradigms: kernel methods, support vector machines, Bayesian inference
- Neural networks are considered "messy" with non-convex optimization landscapes
- Theorists have reluctantly approached deep learning due to its complexity
- **Research opportunities exist in:**
 - Double descent theory
 - Energy-based model theory
 - Optimization in non-convex settings
 - High-dimensional probability estimation

Q2: AI in Safety-Critical Industries

Core Question: How do JEPA-style models interface with first-principles engineering models in nuclear applications?

Key Points from Professor LeCun's Response:

- AMII Labs (Metabolism project) targets complex system control
- Initial business focus: control of power plants, potentially nuclear
- Phenomenological models can be trained on real data + synthetic data from simulations
- Models can self-adjust based on prediction discrepancies
- Timeline for reliable nuclear power control models may be lengthy but represents the goal

Timeline of Topics

Phase	Topic	Key Content
1	Introduction	Research focus, hiring, hierarchical JEPA goals
2	Q&A: Math & AI	Courant Institute, double descent, mathematical challenges
3	Energy-Based Models	Contrast functions, avoiding partition function intractability
4	Q&A: Nuclear Industry	World models, SMRs, safety-critical deployment

Tasks and Assignments

No explicit tasks or assignments were found in this lecture.

This was primarily a research discussion and Q&A session rather than a course lecture with homework or projects.

Summary and Conclusions

Professor LeCun's lecture on "World Models and the Future of AI" presents a compelling vision for the next generation of AI systems. The talk highlights several critical mathematical challenges that remain open:

- Theoretical Understanding Needed:** The double descent phenomenon demonstrates that traditional statistical theory fails to explain modern neural network behavior, creating opportunities for mathematicians.
- Energy-Based Models as an Alternative:** Rather than intractable probabilistic modeling, energy-based contrast functions offer a practical approach to capturing variable dependencies in high dimensions.
- Hierarchical JEPA for Complex Systems:** World models trained on phenomenological data—potentially combined with synthetic data from simulations—offer a path toward AI systems capable of planning and control in safety-critical domains like nuclear engineering.
- Mathematics-AI Integration:** The NYU Courant Institute's prominence reflects the growing importance of applied mathematics in AI research, with significant opportunities for mathematicians to contribute to understanding deep learning theory, optimization, and energy-based models.

The lecture emphasizes that the next wave of AI must move beyond token-based processing (like text in language models) to handle high-dimensional, noisy real-world data through world models and hierarchical planning approaches.

Key Terminology

Term	Definition
JEPA	Joint Embedding Predictive Architecture - A hierarchical planning framework
Double Descent	Empirical phenomenon where test error decreases at very high model complexity
Energy-Based Model (EBM)	A contrast function that measures compatibility between input pairs
World Model	Phenomenological model enabling prediction and planning
Digital Twin	Simulation-based replica of a physical system (contrasted with world models)
Stochastic Gradient Descent	Optimization method using random mini-batch sampling
Partition Function	Normalization constant in probabilistic models (often intractable)
SMR	Small Modular Reactors - simplified nuclear power generation units